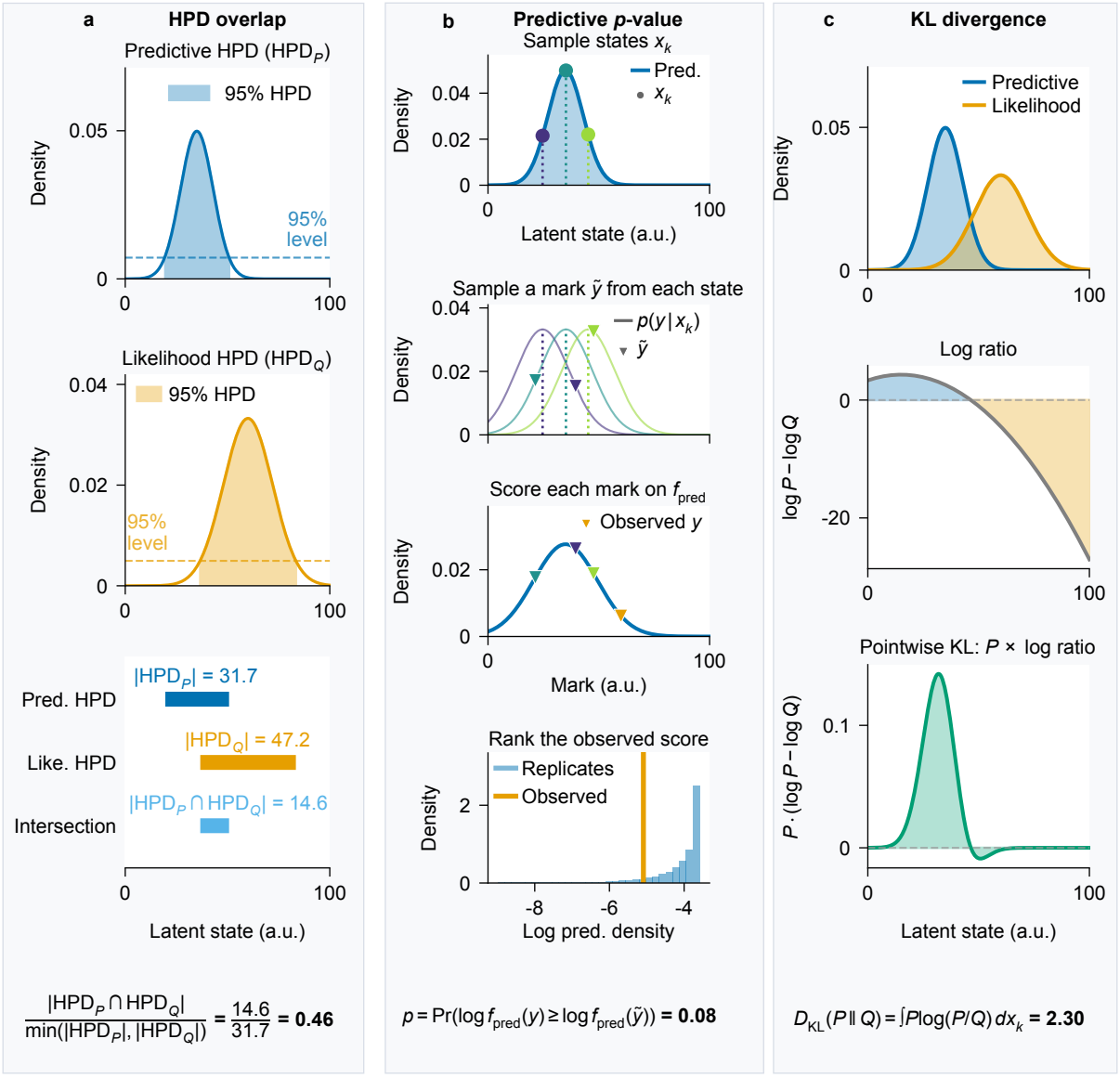

a
HPD overlap
Predictive HPD (HPD_P)
95% HPD
95% level
Density
0.05
0
0
100
Likelihood HPD (HPD_Q)
95% HPD
95% level
Density
0.04
0.02
0
0
100
$|HPD_P|$ = 31.7
Pred. HPD
$|HPD_Q|$ = 47.2
Like. HPD
$|HPD_P \cap HPD_Q|$ = 14.6
Intersection
0
100
Latent state (a.u.)
$\frac{|HPD_P \cap HPD_Q|}{\min(|HPD_P|, |HPD_Q|)} = \frac{14.6}{31.7} = \mathbf{0.46}$
b
Predictive p-value
Sample states x_k
Pred.
x_k
Density
0.04
0.02
0
0
100
Latent state (a.u.)
Sample a mark $\tilde{y}$ from each state
$p(y \mid x_k)$
$\tilde{y}$
Density
0.04
0.02
0
Score each mark on f_{pred}
Observed y
Density
0.02
0
0
100
Mark (a.u.)
Rank the observed score
Replicates
Observed
Density
2
0
-8
-6
-4
Log pred. density
$p = \Pr(\log f_{pred}(y) \geq \log f_{pred}(\tilde{y})) = \mathbf{0.08}$
c
KL divergence
Predictive
Likelihood
Density
0.05
0
0
100
Log ratio
$\log P - \log Q$
0
-20
0
100
Pointwise KL: $P \times$ log ratio
$P \cdot (\log P - \log Q)$
0.1
0
0
100
Latent state (a.u.)
$D_{KL}(P \| Q) = \int P \log(P/Q)\, dx_k = \mathbf{2.30}$